

VIM

Vi IMproved

A Complete Practical Reference to the VIM Text Editor

Modal Editing · Navigation · Visual Mode · Macros · Regex · Customization · Command Reference

1 Chapter · Hands-On Exercises · 7 Command Reference Tables

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Vi IMproved (VIM)

Introduction

Vim, which stands for "Vi Improved", is a highly configurable text editor built to enable efficient text editing. It is an improved version of the vi editor distributed with most UNIX systems. Vim is often called a "programmer's editor," and is so useful for programming that many consider it an entire IDE.

Here are some key features of Vim:

- **Modal Editing:** Vim operates in various modes, with the most commonly used ones being: normal, insert, and visual. Each mode has a different purpose and allows different kinds of commands.
- **Efficiency:** Vim provides a multitude of keyboard shortcuts and commands that allow for fast navigation and editing.
- **Extensibility:** Vim can be customized and extended using scripts and plugins. There are many plugins available that can add extra functionality to Vim.
- **Ubiquity:** Vim is available on almost all modern operating systems, making it a reliable choice for developers.

Install

```
sudo apt update
sudo apt-get -y install vim
```

Simple Exercise

Basic Navigation

- Open a file in Vim.
- Move the cursor to the end of the file using **G**.
- Move the cursor to the beginning of the file using **gg**.
- Move the cursor to the end of the line using **\$**.
- Move the cursor to the beginning of the line using **0**.

Editing Text

- Enter insert mode using **i** and type a sentence.
- Save the changes and exit using **:wq**.
- Open the file again and delete a word using **dw**.
- Undo the change using **u**.
- Redo the change using **Ctrl + r**.

Searching Text

- Search for a word in the file using **/word**.
- Navigate to the next occurrence of the word using **n**.
- Navigate to the previous occurrence of the word using **N**.

Copy and Paste

- Copy a line using `yy`.
- Paste the line below the cursor using `p`.
- Paste the line above the cursor using `P`.

File and Window Operations

- Split the window horizontally using `:split`.
- Split the window vertically using `vsplit`.
- Navigate between windows using `Ctrl + w` followed by the arrow key.

Advanced Editing

- Replace a word using `cw` and then type the new word.
- Replace a character under the cursor using `r` and then type the new character.
- Replace the entire line using `cc` and then type the new line.

Visual Mode

- Enter visual mode using `v`.
- Select some text using the arrow keys.
- Copy the selected text using `y`.
- Paste the copied text using `p`.

Buffers and Tabs

- Open a new file in a new buffer using `:e filename`.
- Switch between buffers using `:bn` and `:bp`.
- Open a new file in a new tab using `:tabe filename`.
- Switch between tabs using `gt` and `gT`.

Macros

- Start recording a macro using `q` followed by a letter to name the macro.
- Perform some actions.
- Stop recording using `q`.
- Play the macro using `@` followed by the name of the macro.

Customizing Vim

- Open your vimrc file using `:e ~/.vimrc`.
- Add a new setting, for example, `set number` to display line numbers.
- Save and exit using `wq`.
- Reload Vim or your vimrc file using `:source ~/.vimrc`.

Block Selection and Editing

- Enter visual block mode using `Ctrl + v`.
- Select a block of text using the arrow keys.
- Change the selected text using `c`, then type the new text and hit `Esc`.

Regular Expressions

- Search for a pattern in the file using `/pattern`.
- Replace the pattern using `:%s/pattern/replacement/g`.

Folding

- Create a fold using `zf` followed by a motion command, like `zf2j` to fold the next two lines.
- Open a fold using `zo`.
- Close a fold using `zc`.

Command-line Mode

- Enter command-line mode using `:`.
- Execute a shell command using `!:command`.

Marks

- Set a mark using `m` followed by a letter.
- Jump to a mark using `'` followed by the letter.

Spell Checking

- Enable spell checking using `:set spell`.
- Navigate to the next misspelled word using `]s`.
- Navigate to the previous misspelled word using `[s`.
- Suggest corrections for a misspelled word using `z=`.

Study Materials — Command Reference Tables

The following tables provide a comprehensive reference for all essential VIM commands. Command keys are shown in monospace; descriptions explain their effect.

Cursor Movement

Command	Description
h	Move left
j	Move down
k	Move up
l	Move right
w	Jump by start of words (punctuation considered words)
W	Jump by words (spaces separate words)
e	Jump to end of words (punctuation considered words)
E	Jump to end of words (no punctuation)

Command	Description
b	Jump backward by words (punctuation considered words)
B	Jump backward by words (no punctuation)
0	(zero) Start of line
^	First non-blank character of line
\$	End of line
G	Go to command (prefix with number, e.g. 5G goes to line 5)

 Prefix any cursor movement command with a number to repeat it. For example, `4j` moves down 4 lines.

Insert Mode — Inserting / Appending Text

Command	Description
i	Start insert mode at cursor
I	Insert at the beginning of the line
a	Append after the cursor
A	Append at the end of the line
o	Open (append) blank line below current line
O	Open blank line above current line
ea	Append at end of word
Esc	Exit insert mode

Editing

Command	Description
r	Replace a single character (does not use insert mode)
J	Join line below to the current one
cc	Change (replace) an entire line
cw	Change (replace) to the end of word
c\$	Change (replace) to the end of line
s	Delete character at cursor and substitute text

Command	Description
S	Delete line at cursor and substitute text (same as cc)
xp	Transpose two letters (delete and paste)
u	Undo
.	Repeat last command

Marking Text (Visual Mode)

Command	Description
v	Start visual mode — mark lines, then do command (e.g. y to yank)
V	Start linewise visual mode
o	Move to other end of marked area
Ctrl+v	Start visual block mode
O	Move to other corner of block
aw	Mark a word
ab	Mark a () block (with braces)
aB	Mark a {} block (with brackets)
ib	Inner () block
iB	Inner {} block
Esc	Exit visual mode

Visual Commands and Cut / Paste

Command	Description
>	Shift right
<	Shift left
y	Yank (copy) marked text
d	Delete marked text
~	Switch case
yy	Yank (copy) a line
2yy	Yank 2 lines
yw	Yank word

Command	Description
y\$	Yank to end of line
p	Put (paste) the clipboard after cursor
P	Put (paste) before cursor
dd	Delete (cut) a line
dw	Delete (cut) the current word
x	Delete (cut) current character

Exiting

Command	Description
:w	Write (save) the file, but don't exit
:wq	Write (save) and quit
:q	Quit (fails if there are unsaved changes)
:q!	Quit and throw away all changes

Search and Replace

Command	Description
/pattern	Search forward for pattern
?pattern	Search backward for pattern
n	Repeat search in same direction
N	Repeat search in opposite direction
:%s/old/new/g	Replace all occurrences of old with new throughout file
:%s/old/new/gc	Replace all occurrences with confirmation prompt

Working with Multiple Files

Command	Description
:e filename	Edit a file in a new buffer
:bnext / :bn	Go to next buffer

Command	Description
:bprev / :bp	Go to previous buffer
:bd	Delete a buffer (close a file)
:sp filename	Open a file in a new buffer and split window horizontally
Ctrl+ws	Split windows horizontally
Ctrl+ww	Switch between windows
Ctrl+wq	Quit a window
Ctrl+wv	Split windows vertically

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